Bridging Communication Gaps for the Hearing Impaired

Project Guide:

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Introduction

- Sign language enables communication for millions with hearing and speech impairments, combining hand gestures and movements to express thoughts and emotions.
- Human interpreters and assistive devices are often costly or inaccessible, creating barriers between signers and non-signers.
- This project translates 282 ASL gestures into text and multilingual speech, promoting inclusivity.



Problem Statement

- Recognizing dynamic ASL gestures is challenging due to variations in speed, lighting, and individual signing styles.
- Users struggle with costly hardware and limited systems that only recognize letters, not full words, restricting effective real-time sign language communication.
- Goal: Develop a cost-effective, webcam-based solution with high accuracy and diverse outputs.

Objective

- Enable real-time gesture recognition using standard webcam input.
- Develop a user-friendly GUI with realtime prediction
- Ensure cost-effectiveness, scalability, and accessibility for diverse users.

Literature Survey

S.No	Title,	Publication/Year	Problem Addressed	Proposed Solution	Pros	Cons
1.	Silent No More: A Comprehensive Review of Artificial Intelligence, Deep Learning, and Machine Learning in Facilitating Deaf and Mute Communication	Springer/2024	This paper looks at how AI can help with solutions like sign language recognition and speech-to-text systems.	Used CNN for static gestures (alphabets, numbers) and LSTM for dynamic gestures, with skin-segmented images for training.	• Leverages AI techniques for accurate real-time solutions.	• Limited datasets used for training and **kinect based**
2.	Bi-Directional Communication Between the Deaf and Dumb People and Normal People	IEEE/2024	Deaf and Dumb people face difficulties communicating with others because most people don't understand sign language.	Used CNN for static gestures (alphabets, numbers) and LSTM for dynamic gestures, with skin-segmented images for training.	• Accurate image classification using CNN and time-based data prediction using LSTM.	• Focused only on Indian Sign Language and **require devices like gloves for gesture** classification

Literature Survey

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3.	Development of Sign Language Translator for Disable People in Two-Ways Communication	IEEE/2023	Many deaf and mute individuals face communication barriers with those unfamiliar with sign language. This creates difficulties in daily interactions and restricts accessibility.	The paper proposes a Python-based system that uses a camera to recognize hand gestures, converting them into text and speech for two-way communication.	• High accuracy, user-friendly, two-way communication, and compatibility with various devices.	• Limited to English and predefined gestures; does not support real-time multi-lingual integration.
4.	SIGN-UNITE ASL Detection using ML for Deaf Mute People Communication	IEEE/2022	Difficulty in communication between signers and non-signers.	Custom CNN for recognizing signs from video frames using the MNIST dataset.	• Strong progress in gesture recognition using deep learning and computer vision techniques.	• Dataset focused; lacks real-time interaction.

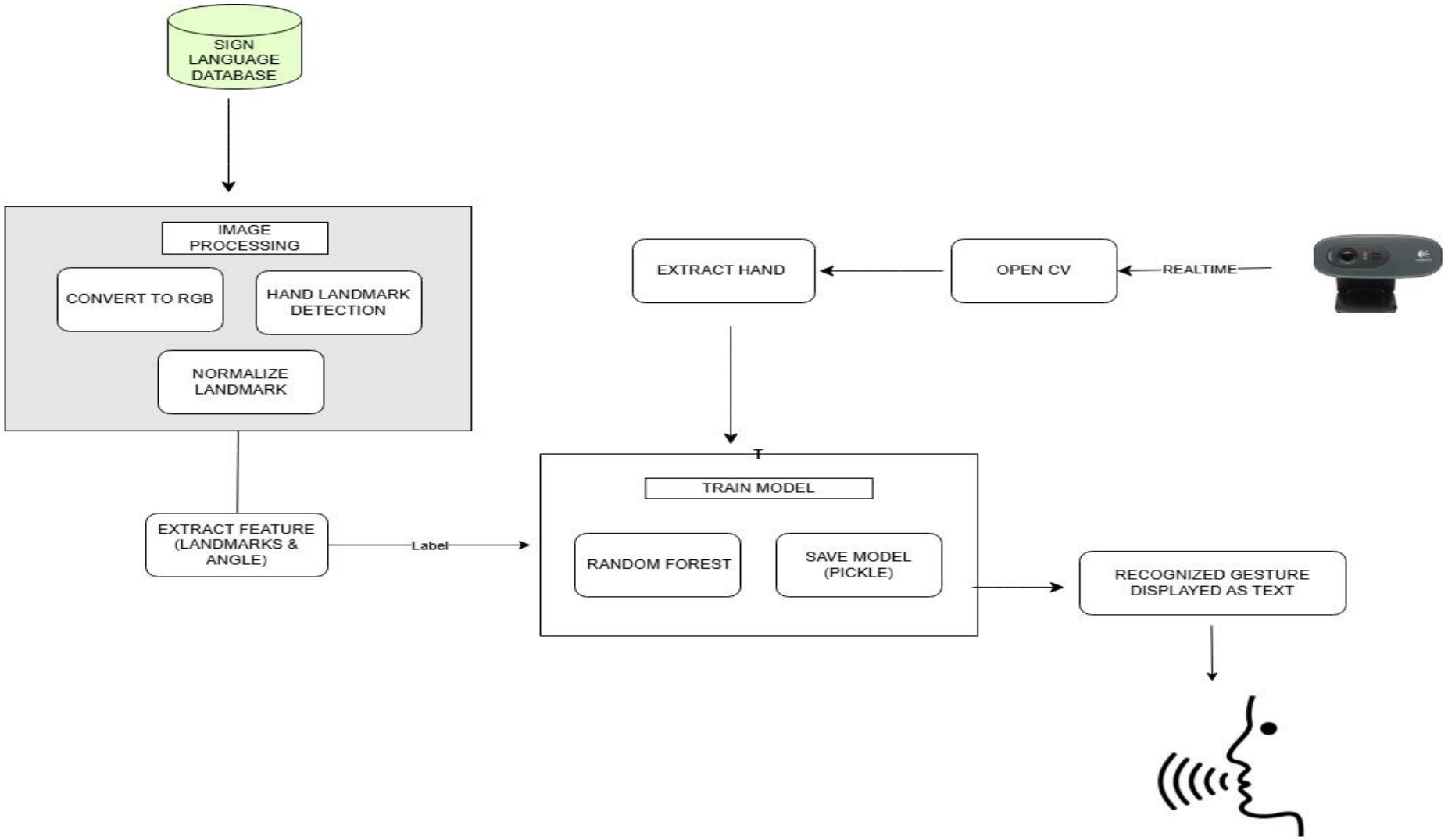
Challenges Addressed

- Existing systems often depend on expensive hardware components, like sensor-equipped gloves or Kinect devices, to achieve real-time operation.
- Many approaches primarily rely on image or video data for gesture recognition instead of other modalities.
- Gesture classification in most systems mainly focuses on recognizing letters and alphabets rather than words.

Proposed System

- Translates 282 ASL gestures into text using a webcam.
- Integrates MediaPipe (hand tracking), Random Forest (classification), and Tkinter GUI (interaction).
- Focuses on real-time performance, scalability, cost effectiveness and user accessibility.

System Architecture



System Architecture

- Input: 640×480 realtime video captured using webcam.
- Processing: MediaPipe extracts 21 keypoints per hand, producing 294 features per frame.

System Architecture

- Classification: Random Forest with 100 trees and max depth 20 processes sequences of 30 frames.

- Output: Recognized text displayed via Tkinter GUI; audio generated in English and Tamil using gTTS and Pygame.

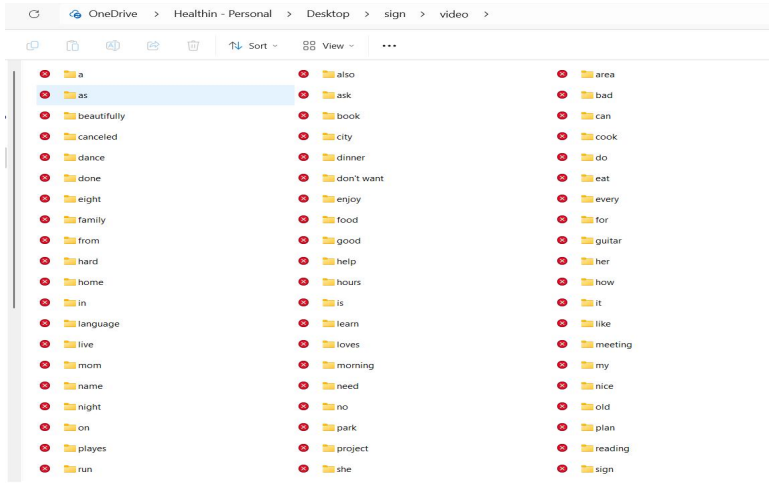
  pygame.mixer.music

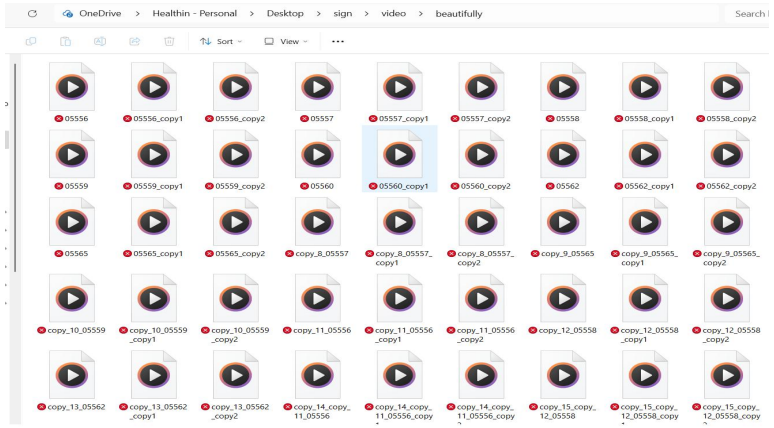
Dataset

- Dataset: WLASL with 282 gestures.
  Train: 13,319 samples.
  Test: 3,330 samples.

- Includes single or double hand gestures

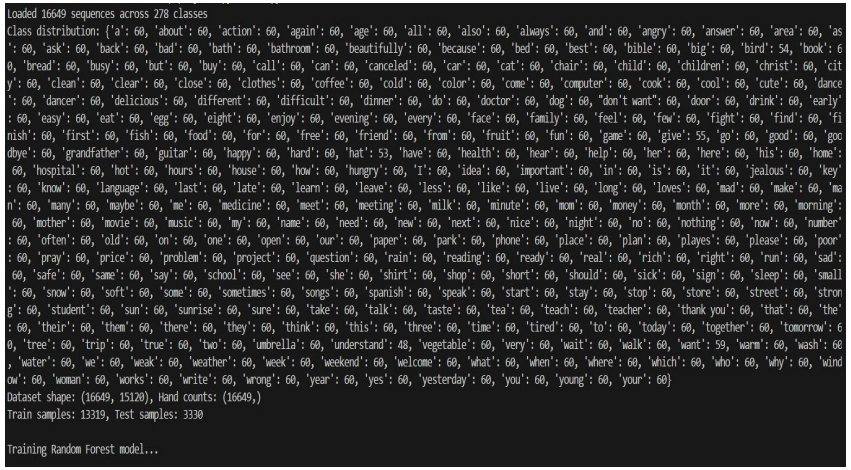
- Stored as MP4 videos, processed into 30-frame sequences.





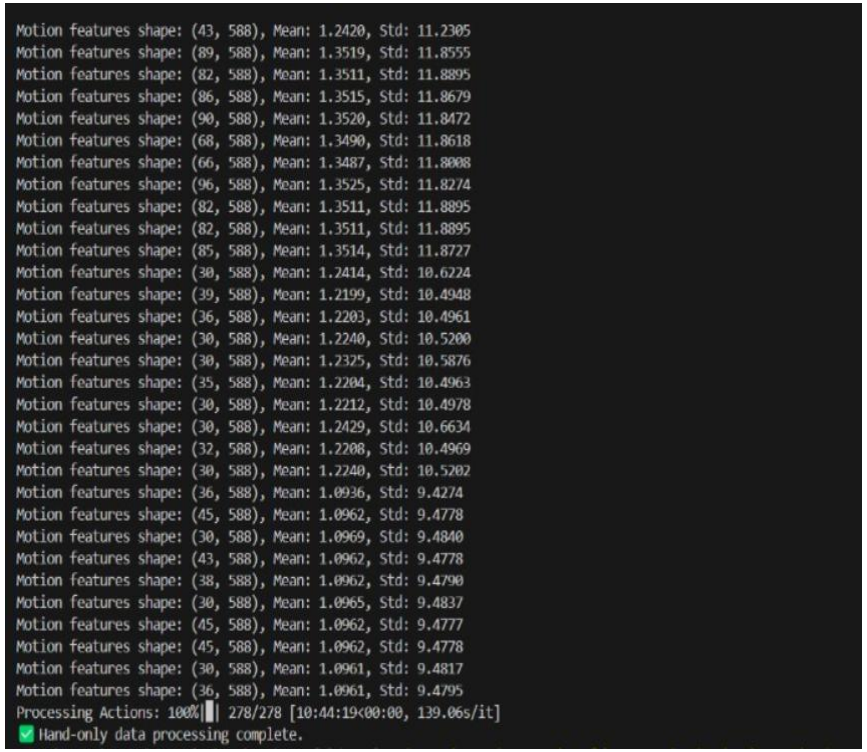
Dataset Preprocessing

- Videos processed frame-by-frame using OpenCV and MediaPipe.
- MediaPipe detects up to two hands, extracting 21 keypoints per hand (x, y, z).
- Keypoints normalized relative to wrist, scaled by hand size to reduce variability.
- Confidence threshold: 0.8 for detection and tracking.



Dataset Preprocessing

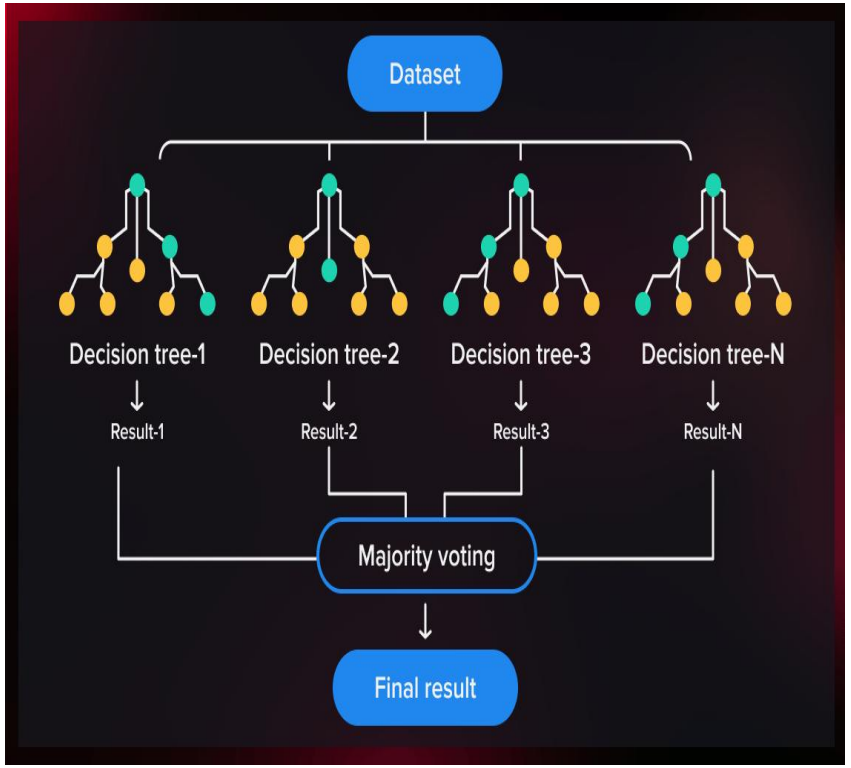
- Features: 126 spatial (angles, distances, palm size), 126 motion (velocity).
- Sequences padded to 30 frames for uniformity.
- Features saved as NumPy arrays for training and real-time prediction.



Random Forest Classifier

- Random Forest: 100 trees, max depth 20, trained on 7,560-dimensional vectors.

- It takes flattened motion-based feature vectors (position + velocity) extracted from 30-frame sequences of hand landmarks.



Random Forest Classifier

- Multiple decision trees are trained on random subsets of the data; each tree makes an independent prediction.
- The final gesture label is chosen based on the majority vote from all the trees in the forest.
- Predictions are accepted only if their confidence exceeds a defined threshold to ensure accuracy and stability.

Algorithm for Feature Extraction

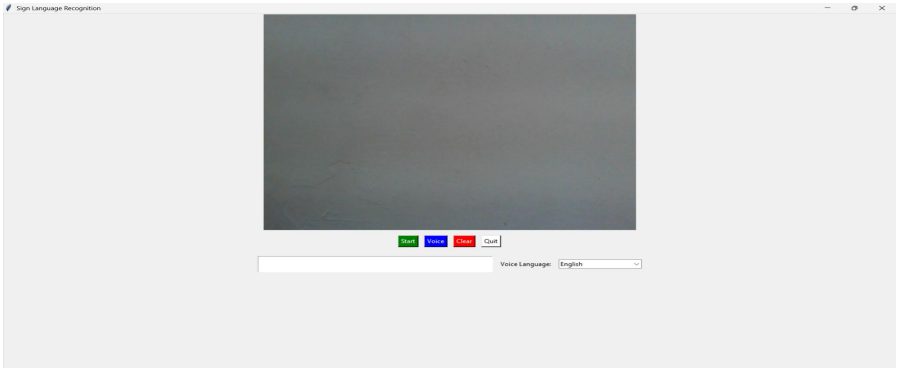
```
Input  : Video frame frame (RGB image)
Output: features (252-dimensional vector), hand_count
ExtractFeatures(frame) features ← zeros(252);
hand_count ← 0;
Convert frame from BGR to RGB;
Detect hands using MediaPipe → results_hands;
if results_hands contains hand landmarks then
    hand_count ← number of detected hands (max 2);
    foreach hand h in results_hands up to 2 do
        Extract 21 keypoints (x, y, z);
        Normalize keypoints relative to wrist;
        Scale by hand size (wrist to index tip);
        Compute spatial features (63 values);
        Compute geometric features (angles, lengths, etc.);
        Append to features;
    if previous frame exists then
        Compute velocity = difference from previous frame;
        Append velocity to features total: 252
else
    return zeros(252), 0;
return features, hand_count;
```

Algorithm for Gesture Classification

```
Input  : Sequence seq (30×252), Previous word prev_word, Hand coun
         hand_count
Output: Predicted gesture, max_proba (confidence)
PredictGesture(seq, prev_word, hand_count) rf_model ←
LoadModel();
flat_seq ← Flatten(seq);
proba ← rf_model.predict_proba(flat_seq);
if prev_word is empty then
 ⌊ allowed_words ← STARTER_WORDS;
else
 ⌊ allowed_words ← NextWordFromNgram(prev_word);
if hand_count = 1 then
 ⌊ allowed_words ← SINGLE_HAND_ACTIONS;
else if hand_count = 2 then
 ⌊ allowed_words ← DOUBLE_HAND_ACTIONS;
proba ← FilterProba(proba, allowed_words);
max_proba ← max(proba);
if max_proba ≥ 0.011 then
 ⌊ gesture ← GetLabelFromActions(proba);
else
 ⌊ gesture ← null;
return gesture, max_proba;
```

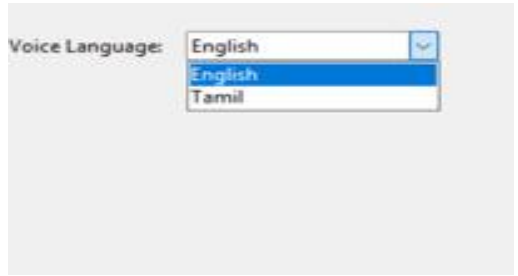
Implementation (GUI)

- Tkinter GUI integrates:
  - OpenCV: Live video feed.
  - MediaPipe: Hand landmark visualization.
  - Pygame/gTTS: Audio output.



- Features: Control buttons (Start/Stop, Voice, Clear, Quit), text box for predictions, language dropdown.





- Updates every 10ms for responsiveness.

Gesture Classification Report

- Classification Report: High F1-scores (e.g., 0.96 for “yes”); lower for “your” (0.73).

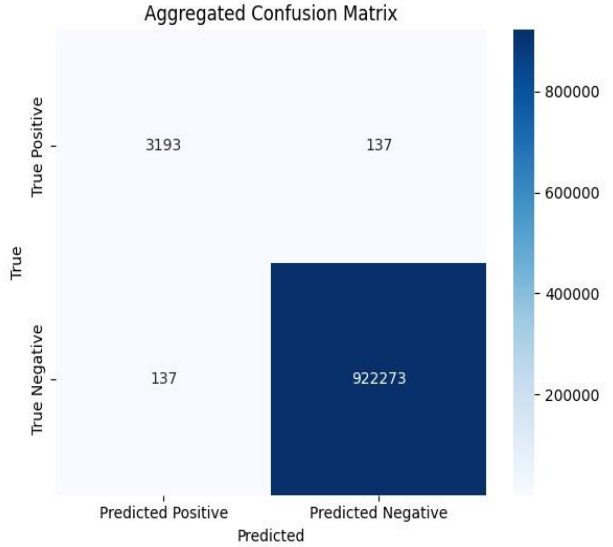
Gesture	Precision	Recall	F1-Score	Support
a	0.92	1.00	0.96	12
about	1.00	1.00	1.00	12
action	0.92	1.00	0.96	12
age	0.86	1.00	0.92	12
all	0.92	1.00	0.96	12
also	0.83	0.83	0.83	12
always	1.00	1.00	1.00	12
and	0.92	1.00	0.96	12
angry	1.00	1.00	1.00	12
answer	1.00	1.00	1.00	12
⋮	⋮	⋮	⋮	⋮
yes	1.00	0.92	0.96	12
you	1.00	0.92	0.96	12
your	0.80	0.67	0.73	12
Accuracy	**0.95**	**0.95**	**0.95**	**3330**
Macro Avg	0.95	0.95	0.95	3330
Weighted Avg	0.95	0.95	0.95	3330

Evaluation Metrics

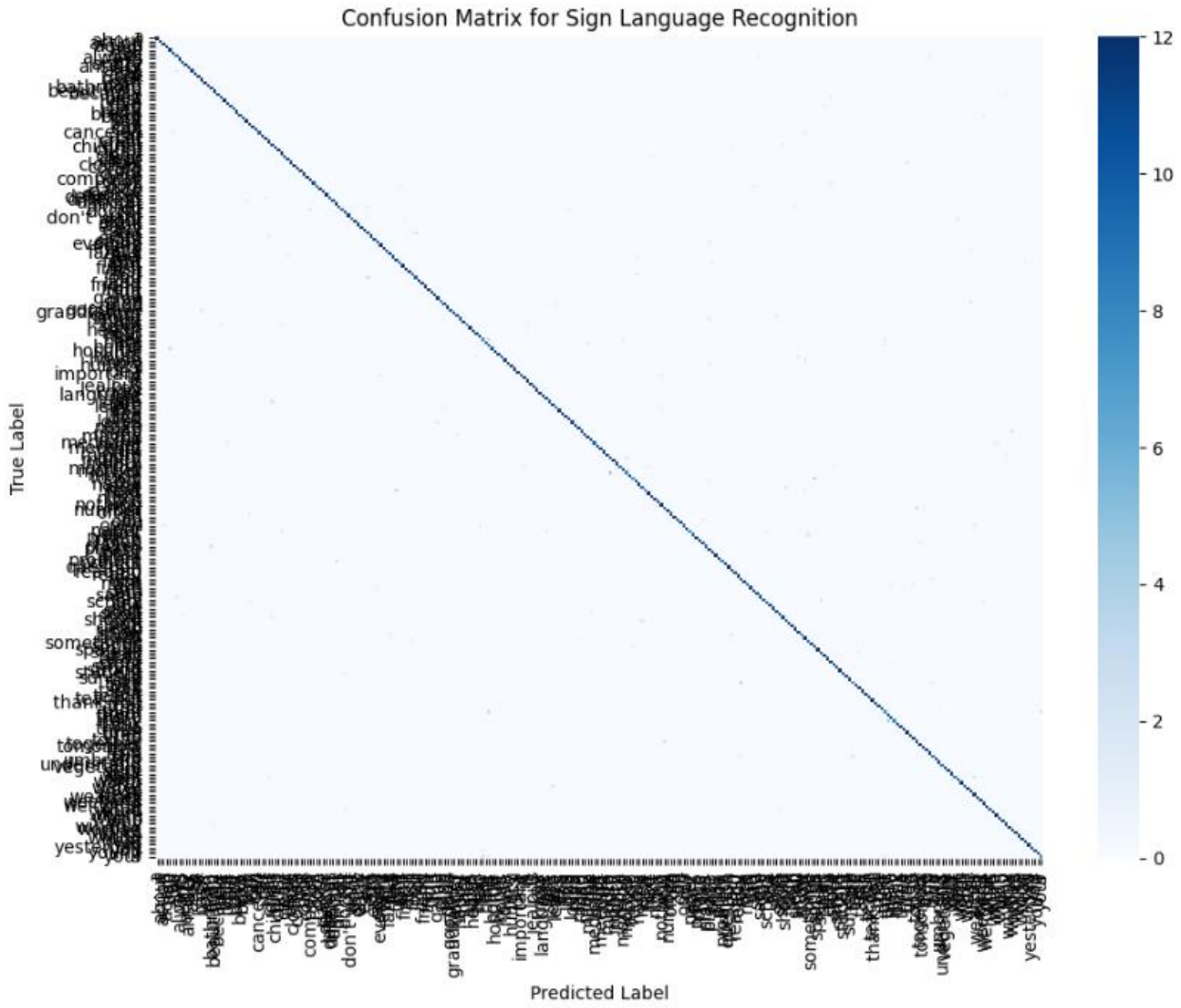
- Accuracy: 94.77% on 3,330 test samples.
- Macro F1-Score: 94.73%
- Confusion Matrix:
  - ➢ True Positives: 3,193.
  - ➢ True Negatives: 922,273.
  - ➢ False Positives/Negatives: 137 each.

```
Training Random Forest model...

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Random Forest Model Evaluation
Accuracy: 94.77%
Macro F1: 94.73%
```



Confusion Matrix



Tools and Technologies

- OpenCV: Video capture/processing.
- MediaPipe: Hand landmark detection.
- Scikit-learn: Random Forest classifier.
- Tkinter: GUI development.
- gTTS/Pygame: Multilingual audio.
- Google Translate API: Text translation.

Conclusion & Future Work

Conclusion:

- Built a real-time ASL recognition system with 91.5% accuracy using MediaPipe and Random Forest.
- Cost-effective, scalable, user-friendly.
- Supports 282 gestures.

Future Work:

- Expand gestures, add facial/body cues, and reduce latency.
- Support mobile deployment and smart device integration.

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THANK YOU